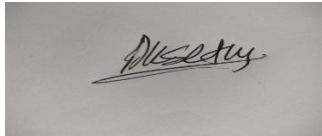
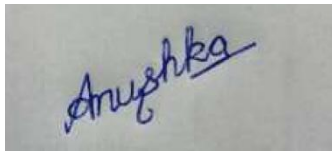
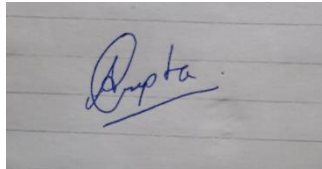


NourishShare

Autonomous Food Delivery System Using Real-Time Geospatial
Matching for Contactless and Efficient Last-Mile Delivery

APPLICANT(S)

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PREAMBLE TO DESCRIPTION

COMPLETE SPECIFICATION –

The following specification particularly describes the invention and the manner in which it is to be performed.

TITLE:

NourishShare: Autonomous Food Delivery System Using Real-Time Geospatial Matching for Contactless and Efficient Last-Mile Delivery

FIELD OF THE INVENTION

[0001] The current invention is in the domain of web-based systems and services, geospatial systems and intelligent resource allocation systems, in particular automated food redistribution systems. The invention comprises software-mediated coordination and real-time data analytics. This facilitates the allocation of surplus food in urban areas.

[0002] In particular, the invention relates to a real-time geospatial food delivery and redistribution matching system for data-driven decision-making, in which the system combines location-based services, user interface and matching algorithms. The system uses spatial and temporal information to facilitate allocation. This leads to effective coordination between suppliers and demanders of food.

[0003] The system facilitates autonomous matching and coordination of food delivery, contactless operation, and resource allocation in indoor and outdoor settings. It allows automated matching and decision-making to reduce human involvement. This enhances operational efficiency, decreases response time, and increases the system's reliability.

BACKGROUND

[0004] With the rise in food delivery and growing food waste in cities, there is a pressing demand for effective redistribution networks. A considerable amount of nutritious food is wasted each day while many communities are deprived of resources. This scenario calls for smart and scalable systems.

[0005] Traditional food redistribution networks generally coordinate via phone calls, texts or volunteers. These methods are unstructured and inefficient, resulting in delays. This results in waste due to expired perishable food items.

[0006] Current digital systems offer some solutions, but may lack automated geospatial matching and real-time coordination. They often rely on manual recipient selection, slowing down processes. This restricts their use in urban areas.

[0007] Additionally, some systems do not take temporal factors like food expiry into account. This leads to sub-optimal prioritisation and huge food waste. Temporal considerations must be integrated for redistribution.

[0008] Further, the absence of integration between location-based services and decision making automation mechanisms reduces responsiveness. This affects the responsiveness to dynamic demand and supply. It also has a negative effect on scalability in multi-user scenarios.

[0009] Thus, there is a need for a smart and autonomous food delivery system that can perform real-time geospatial matching, allocation and contactless coordination. This system should take into account spatial and temporal factors. This will enhance food redistribution.

OBJECTS OF THE INVENTION

[0010] To build a self-organising food delivery system that can redistribute surplus food without ongoing management. The system will function with matching algorithms. This allows for efficient and scalable last-mile services.

[0011] To provide real-time spatial matching using location services. Geospatial proximity is used to find recipient organisations nearby. This speeds up the allocation process and shortens the delivery time.

[0012] To provide smart allocation using proximity-based filtering and decision-making. Optimal recipient selection takes place automatically based on criteria. This enhances efficiency and simplifies the process.

[0013] To offer contactless delivery through secure interaction mechanisms. The system enables safe exchange of food items from donors to recipients. This improves safety and customer confidence.

[0014] To reduce delivery time, human labour and cost by automation. The system coordinates activities and minimizes manual effort. This supports large-scale deployment.

[0015] To provide safe and reliable communication between users of the system through user authentication and access control. The system avoids misuse and guarantees data security. This improves overall system reliability.

START



User Registration / Login (Donor / NGO Authentication)



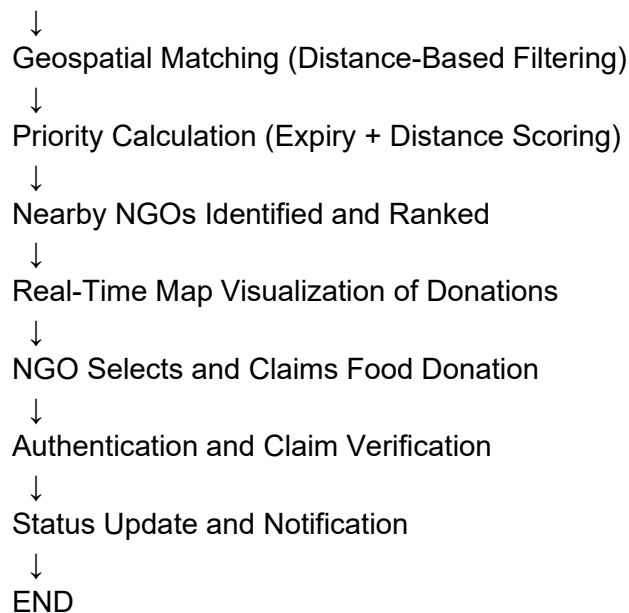
Donor Submits Food Details (Location, Quantity, Expiry Time)



Geocoding Process (Address → Coordinates Conversion)



Data Processing and Storage (Database Handling)



BRIEF DESCRIPTION OF DRAWINGS

[0016] The figure illustrates system flow of the self-organising food redistribution system. This figure shows the workflow from food donation to distribution. The diagram illustrates the interaction between various system components.

[0017] Users input the details of food donations through a web interface: the location, quantity and expiry date. This information is handled by the system. This allows structured data processing.

[0018] The system receives the input data and transforms the location data into geographic information using geocoding. This guarantees the correct mapping of locations. It enables accurate distance measurements.

[0019] It matches in real time to find local recipient organisations. It shows real-time results, prioritizing distance and availability. This ensures efficient allocation.

[0020] Food is prioritised for allocation based on expiry time and distance travelled to reduce waste. Food closer to expiry is prioritised. This improves redistribution effectiveness.

[0021] Results are presented in an interactive map view to help organisations see and accept donations. The system also provides real-time status. This improves user interface and visibility.

DETAILED DESCRIPTION OF THE INVENTION

[0022] The invention is a web system with a client-server model, using RESTful APIs for the frontend interface to interact with the back-end services. This allows for high-speed data transfer and interaction. The system is scalable and modular.

[0023] The system has a frontend module for users to register, provide information about their food donations and engage with the system. The user interface is responsive and intuitive. This enhances user experience for various users.

[0024] The backend system receives user input, manages the flow of data and runs matching algorithms. It performs essential system operations like user authentication, data processing and matching. This ensures reliable system performance.

[0025] The system is linked to geospatial services to transform location details from text into coordinates using geocoding. This facilitates precise location identification. It also enables proximity-based searches.

[0026] A proximity-matching algorithm is used to select recipient organisations within a certain geographical distance. The algorithm uses spatial data to calculate distance. This reduces transportation time.

[0027] The system uses first-in-first-out system, prioritising allocation of food items with a shorter lifespan. This reduces wastage of perishable products. It improves allocation efficiency.

[0028] It has a safe storage compartment to transport food products. The compartment is designed to be stable and prevent leakage. This guarantees safe and clean delivery of goods. A ranking system uses distance and expiry to rank potential matches. Each potential match is scored. This allows for smart matching.

[0029] The system includes a user authentication system to prevent any unauthorised access to the delivery compartment. This can be achieved through OTP, QR code or mobile app verification. This will prevent tampering and ensure delivery to the right place. Our system offers real-time mapping using a map view. Users are able to see nearby donations and accept them immediately. This improves decision-making and user experience.

[0030] The system is able to connect with IoT and cloud computing for added features. This allows real-time monitoring, data sharing and remote operation of the robot. This enhances system performance and operation. Strong authentication measures like token-based authentication are in place. This guarantees only authenticated users have access. This improves system security and dependability.

[0031] Cloud connectivity enables the system to provide route planning, monitoring and management. It can control and manage multiple robots. This allows the system to be scaled up. The system can run real-time updates and multi-users. Several donors and organisations can access the system. This ensures scalability and performance.

[0032] The modular design enables the platform to be integrated with other delivery systems, such as autonomous delivery in the future. This enhances flexibility and adaptability. It supports future technological advancements.

CLAIMS

We Claim:

1. A self-operating food distribution system, which includes an online platform, a geospatial matching engine, and a data management system for real-time matching of food donors with recipient organizations. This system is intended to receive user input and facilitate automated coordination of food redistribution programs.
2. The system as claimed in claim 1, wherein the system uses geocoding to translate location information into coordinates for geographic mapping. Geocoding allows for accurate mapping and spatial calculations.
3. The system as claimed in claim 1, wherein the system conducts geospatial matching in real time based on proximity. Matching dynamically searches for organisations within a certain geographic proximity.
4. The system as claimed in claim 1, wherein the system matches food donations based on system-defined criteria. The system automates the allocation process by choosing the best recipient based on the system's recommendations.
5. The system as claimed in claim 1, wherein the system includes prioritization based on expiry dates for effective redistribution of food items. Products with a shorter expiry date are prioritised to reduce food waste.
6. The system as claimed in claim 1, wherein the system employs a spatial-temporal scoring system. The mechanism scores distance and expiry to allocate the food items.
7. The system as claimed in claim 1, wherein the system offers real-time mapping via a mapping interface. It allows users to see food donations and take actions related to closeness and availability.
8. The system as claimed in claim 1, wherein the system adopts secure authentication and user access. The system provides secure access to data for claiming, managing or viewing food donations.
9. The system as claimed in claim 1, wherein the system is capable of multi-user interaction. The system supports multiple donors and recipients without compromising system performance.
10. The system substantially as described herein offers a smart contactless food delivery system. The system optimises processes, minimises manual administration, and enables easy access to excess food.

ABSTRACT

The current invention provides an autonomous food delivery system to enhance the effectiveness of food redistribution by incorporating real-time geospatial processing. This system combines web technologies, geo-location services, and smart matching algorithms to match food donors with local recipients. It is designed to work via a web platform that facilitates communication between various parties, enabling better coordination and access.

The system employs geocoding and proximity filtering to efficiently match potential recipients with appropriate locations. Moreover, it has expiry-based processing to prioritize the timely redistribution of perishable food products, thus reducing food waste. An interactive real-time map provides visibility of available food donations, location tracking and claim operations, improving transparency and informed decision making.

The system also involves a secure user authentication system and a scalable system that supports multiple users and simultaneous operations. It also dynamically updates matching and allocation based on real-time data inputs. By eliminating administrative overheads, avoiding delays, and efficiently allocating resources through smart automation, the invention offers an efficient and scalable approach to tackling food wastage and enhancing food security in a city.

Moreover, the system's modular architecture enables future expansion with technologies such as predictive analytics, optimal route planning and automated delivery systems. This allows for flexibility to meet changing needs and can be deployed in large social resource distribution systems and smart cities.